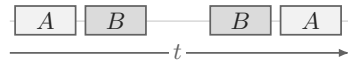


One recording x

$A = \text{dog}, B = \text{rooster}$



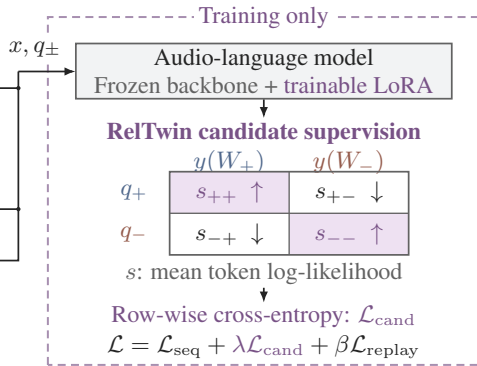
$W_+ : A \rightarrow B$ $W_- : B \rightarrow A$

q_+ : dog followed by rooster

q_- : rooster followed by dog

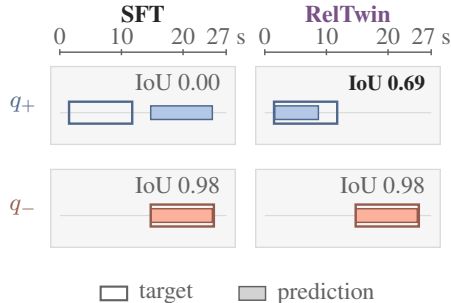
Both relations hold in x ;
each query has a different target.

(a) Inverse-relation pairs



(b) Paired answer supervision

Dev example 10007-v0; seed 0



(c) Query-specific localization